

Academic Year: ( 2026 / 2027 )

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Department assigned to the subject: Continuum Mechanics and Structural Analysis Department

Coordinating teacher: ARTERO GUERRERO, JOSE ALFONSO

Type: Compulsory ECTS Credits : 6.0

Year : 3 Semester : 1

**REQUIREMENTS (SUBJECTS THAT ARE ASSUMED TO BE KNOWN)**

We strongly advise you not to take this course if you have not passed Physics I , Mechanics Applied to Aerospace Engineering and Introduction to structural analysis.

**LEARNING OUTCOMES**

CB2: Students are able to apply their knowledge to their work or vocation in a professional manner and possess the competences usually demonstrated through the development and defence of arguments and problem solving within their field of study.

CB5: Students will have developed the learning skills necessary to undertake further study with a high degree of autonomy.

CG2: Planning, drafting, direction and management of projects, calculation and manufacturing in the field of aerospace engineering.

CG5: Ability to carry out projection activities, technical management, expert appraisal, drafting of reports, opinions, and technical advice in tasks related to Aeronautical Technical Engineering, the exercise of genuinely aerospace technical functions and positions.

CG9: Ability to analyse and solve aerospace problems in new or unknown environments, within broad and complex contexts, integrated in multidisciplinary and international work teams.

CG10: Ability to use computational and experimental tools for the analysis and quantification of engineering problems.

CE.CRA1: Understand the behaviour of structures under stresses in service conditions and limit situations.

CE.CRA9: Adequate knowledge and applied to Engineering of: The principles of the mechanics of the continuous medium and the techniques for calculating its response.

CE.CRA13: Applied knowledge of: the science and technology of materials; mechanics and thermodynamics; fluid mechanics; aerodynamics and flight mechanics; navigation and air traffic systems; aerospace technology; theory of structures; air transport; economics and production; projects; environmental impact.

RA1: Have basic knowledge and understanding of mathematics, basic sciences, and engineering within the aerospace field, including: behaviour of structures; thermodynamic cycles and fluid mechanics; the air navigation system, air traffic, and coordination with other means of transport; aerodynamic forces; flight dynamics; materials for aerospace use; manufacturing processes; airport infrastructures and buildings. In addition to a specific knowledge and understanding of the specific aircraft and aero-engine technologies in each of the subjects included in this degree.

RA2: Be able to identify aerospace engineering problems, recognise specifications, collect and interpret data and information, establish different resolution methods and select the most appropriate among the available alternatives.

RA3: Be able to carry out designs in the field of aerospace vehicles, propulsion systems, navigation and air traffic control, airport infrastructures, or equipment and materials for aerospace use, which comply with the required specifications, collaborating with other engineers and graduates.

RA4: Graduates will be able to carry out initial research methods approaches commensurate with their level of knowledge involving literature searches, design and execution of experiments, data interpretation, selection of the best proposal and computer simulation.

RA6: Have the necessary skills for the practice of engineering in today's society.

**OBJECTIVES**

Knowledge of the basic tools for the calculation of thin-walled beams, that provide to the student the ability to design structural components of the aerospace industry.

Acquisition of the technological knowledge needed to calculate bidimensional structural elements used in aerospace structures.

Knowledge of the basics of the design of structures made of composite materials, including composite laminates and sandwich structures, which are widely used in aerospace industry.

Familiarity with the fundamentals of the design of the main structural elements and systems used in aircrafts.

Ability to use specific software to analyse, design and calculation of structural elements, developing a critical awareness.

## DESCRIPTION OF CONTENTS: PROGRAMME

### Chapter 1. Structures in the aerospace and aeronautical sector

#### Subject 1. Structural description of the aircraft

- 1.1 Loads on aircraft structures
- 1.2 Function of structural components
- 1.3 Wing structure
- 1.4 Fuselage structure
- 1.5 Stabilizers structure
- 1.6 Helicopter structure

#### Subject 2. Structures in the aeronautical sector

- 2.1 Frame and truss structures
- 2.2 Space structures
- 2.3. Future trends
- 2.4. Energetic methods
- 2.5. Thermal loads

### Chapter 2. Bending, shear and torsion of thin-walled beams

#### Subject 3 and 4. Bending and shear of open and closed, thin-walled beams

- 3.1 Kinematic hypothesis
- 3.2 Shear of open section beams
- 3.3 Shear of closed section beams
- 3.4 Shear centre

#### Subject 5. Torsion of beams

- 5.1 Torsion of closed section beams
- 5.2 Torsion of open section beams

#### Subject 6. Torsion on multiple-cell thin-walled beams

- 6.1 Torsion of multiple-cell closed section beams
- 6.2 Torsion of multiple-cell open section beams

### Chapter 3. Plates and Shells

#### Subject 7 and 8. Bending of thin plates

- 7. 1 Kinematic
- 7.2 Plates subjected to a distributed transverse loads
- 7.3 Plates subjected to bending and twisting
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#### Subject 9 and 10. Shells

- 9.1 Hypotheses
- 9.2 Thin shells subjected to in-plane loads
- 9.3 Thin shells subjected to bending loads

### Chapter 4. Laminate and sandwich structures

#### Subject 11. Theory of laminate

- 11.1 Kinematic
- 11.2 Orthotropic constitutive equations
- 11.3 Classical and first-order theories of laminate composites
- 11.4 Failure criteria

#### Subject 12. Composite beams and plates

- 12.1 Composite beams subjected to bending
- 12.2 Composite thin-walled cross-section beams
- 12.3 Bending of composite plates

### Subject 13. Sandwich structures

- 13.1 Basic sandwich theory
- 13.2 Sandwich beams
- 13.3 Sandwich plates

## LEARNING ACTIVITIES AND METHODOLOGY

The Learning Activities are (approximate hours for each are indicated in parentheses)

AF1: Theoretical-Practical Sessions (44 hours)

AF2: Workshops and Laboratories (6 hours)

AF3: Tutorials (4 hours)

AF4: Independent or Group Work by the Students (98 hours)

AF9: Final Exam (4 hours)

The Teaching Methodologies are

MD1: Lectures

MD2: Practical Sessions

MD3: Laboratory Sessions

MD4: Tutorials

In each week one lecture session (master class) and one practical session (in reduced groups) will be taught. The first is geared to the acquisition of theoretical knowledge, and the second to the acquisition of practical skills related to theoretical concepts. Additionally, students will complement the classes with work at home, using material provided on Aula Global.

In addition to these sessions, four laboratory practical sessions in reduced groups (maximum 20 students) will be impart. These practices are mandatory.

Students also have the possibility of individual tutorials.

## ASSESSMENT SYSTEM

**% end-of-term-examination/test:** 60

**% of continuous assessment (assignments, laboratory, practicals...):** 40

Final exam (mandatory): 60%

Continuum assessment: 40%

- Laboratory report: 15%

- Evaluation tests: 15%

- Topic report: 10%

If the mark obtained in the final exam is lower than 4.5, the final mark of the student will be computed only with the final exam.

In order to pass the course, the attendance and performance of the laboratory practices foreseen in the weekly planning are compulsory. The weighting of the laboratory practice mark in the continuous assessment corresponds to what it is established in the course, in accordance with the regulations of the university. In the subject Aerospace structures, the weighting of the laboratory practices takes the value of 37,5%(\*) of the continuous assessment grade.

Use of Artificial Intelligence tools selectively allowed in this subject. The faculty may indicate a list of works and exercises that the student can perform using AI tools, specifying how they should be used, and how the student should describe the use made of them. If the use of AI by the student gives rise to academic fraud by falsifying the results of an exam or work required to accredit academic performance, the provisions of the Regulation of the University Carlos III of Madrid of partial development of the Law 3/2022, of February 24th, of University Coexistence will be applied.

## BASIC BIBLIOGRAPHY

- Barbero E.J. Introduction to composite materials, Taylor and Francis, 1999

- Megson, T.H.G. Aircraft structures for engineering students, Elsevier, 2007
- Timoshenko, S.P. Theory of plates and shells, McGraw Hill, 1st ed. 1940

#### ADDITIONAL BIBLIOGRAPHY

- Daniels I.M. , Isahi Engineering Mechanics of composite materials, Oxford Univerisity Press, 1994
- Ugural, A. C. Stresses in beams, plates, and shells, Taylor & Francis, 2009
- Vinson, J. R. The Behavior of thin walled structures: beams, plates, and shells, Kluwer Academic Publishers, 1989